



INFOKIT

**LTA SENIOR MANAGEMENT VISIT TO
BEIJING, TIANJIN
CHINA**

24 – 26 JUNE 2026

For DRDG

– Ms Wan Fang

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1. Delegation List

LTA

S/N	Name	Designation
1	Ms WAN Fang 万方	Director, Rail Digitalisation & Guild 铁路数字化和行业协会，处长
2	Ms WANG Yizhen 王易蓁	Manager, Rail Digitalisation & Guild 铁路数字化和行业协会，经理
3	Mr TAY Kiat Chong 郑杰聪	Manager, Rail Digitalisation & Guild 铁路数字化和行业协会，经理
4	Mr LI Ruizhe 李瑞哲	Senior Engineer, Rail Digitalisation & Guild 铁路数字化和行业协会，高级工程师

2. Overview of Itinerary

Date	Main Activities for the Day	Night Stay
24 June Wed	Flight from SG to Beijing PEK <CA976> Departure: 24 Jun, 09:00, Terminal 1 Arrival: 24 Jun 15:15	Beijing
25 June Thu	- Drive to Tianjin - Visit to TCTE Factory and Low-Altitude Economy Sci-Fi Park - Drive back to Beijing - Visit to TCT Headquarter	Beijing
26 June Fri	- Visit to CRRC Academy - Flight from Beijing PEK to SG <CA969> Departure: 26 Jun, 15:45, Terminal 3 Arrival: 26 Jun, 21:55, Terminal 1	

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3. Administrative Details

3.1 Weather Forecast

24 JUN WED	25 JUN THU	26 JUN FRI
Beijing	Tianjin/ Beijing	Beijing
DAY 30 °C 	DAY 30 °C 	DAY 31 °C 
NIGHT 17 °C 	NIGHT 17 °C 	NIGHT 19 °C 

3.2 Hotel Details







Beijing – Muji Hotel Beijing

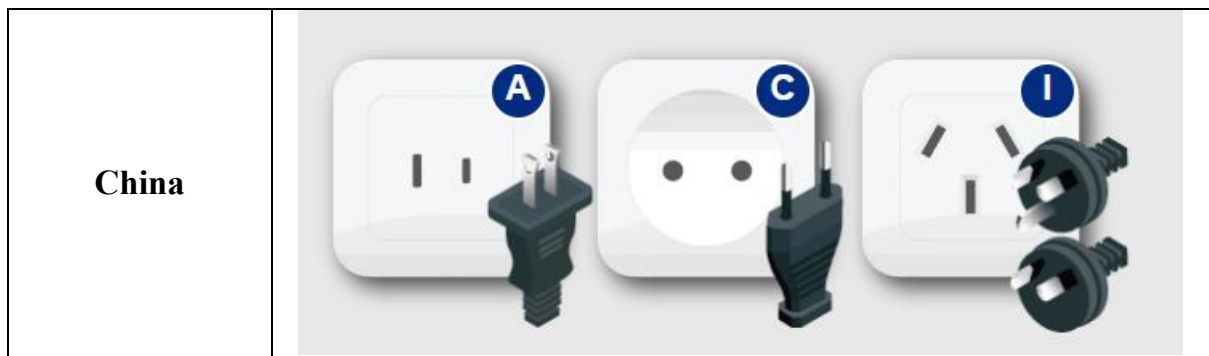
Address: No. 02, Building 2, No. 21 Courtyard, Langfang 1st Lane, Dashilan
Subdistrict, Xicheng District, Beijing, China

Breakfast Hours: 07:30 – 10:00
Muji Café: 11:30 – 20:30
Muji Diner: 07:30 – 22:00

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3.3 Miscellaneous

- a. China has three different types of pins. Travel adapters are recommended.
- b. Dress Code: Business Casual
- c. Remember to bring: Name cards and Passports



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4. Detailed Itinerary

4.1 Day 0 (Wednesday, 24 June 2026, Beijing)

Time	Activity	Remarks
09:00	Flight to Beijing <CA976>	
15:15	Flight Arrives at Beijing PEK	
16:00	Travel to Beijing Hotel	~50 min journey
17:00	Check-in to Hotel	
18:00	Dinner	

4.2 Day 1 (Thursday, 25 June 2026, Beijing & Tianjin)

Time	Activity	Remarks
08:30	Travel to Tianjin (TCT provides transport)	~1h 15min journey
10:00	Visit TCTE & UAV • Introductions • Tour of Intelligent Production Lines • Tour of UAV Scenario Demonstration Visit	
12:30	Lunch and Travel Back to Beijing (TCT provides transport)	~1h 20min journey
14:30	Visit TCT Headquarter • Demonstration of Robot, CdM System and AI use cases • Exchange on Framework Agreement • Group photo	National Engineering Research Center of Rail Transportation Operation and Control Systems
18:30	Dinner with TCT delegates	
20:00	Return to Beijing Hotel (LTA gets didi)	~40 min journey

4.3 Day 2 (Friday, 26 June 2026, Beijing)

Time	Activity	Remarks
08:15	Check-out of Beijing Hotel	
08:30	Travel to CRRC Academy (LTA gets didi)	30min journey
09:00	Discussion with CRRC Academy • Welcome Remarks • Mutual introduction on organisation and key business area • Technical exchange on high-quality dataset development	
11:30	Tour in CRRC Academy	

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	• Visit CRRC Academy laboratories, research platforms, and relevant facilities of learn about achievements in high-quality dataset development, platform capabilities, as well as staff culture initiatives and institutional honors. • Group photo	
12:00	Lunch	
13:00	Travel by Car to Beijing PEK Terminal 3 (LTA gets didi)	~1h journey
14:00	Check in flight	
15:45	Flight back to SG <CA969, reach at 21:55>	

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5. Companies

5.1 Traffic Control Technology



5.1.1 Background

TCT is an overall rail transit solution provider in China, providing the butler service of full life cycle, taking CBTC technology with independent intellectual property rights as the core and specializes in R&D of urban rail transit signalling system and key equipment, system integration, general contracting and maintenance of signalling system and other technical services. Besides, TCT provides urban rail transit with solutions such as communication-based train control system (CBTC), full-automatic operation system (FAO), interoperable communication-based train control system (I-CBTC), urban rail cloud system and TIDS, covering standard establishment, theoretical research, system design, product development, intelligent manufacturing, test and verification, project implementation, training and operation maintenance, so as to realise full life cycle service and guarantee of rail transit signalling system.

5.1.2 Key Representatives

Traffic Control Technology	
	Wen Yilong Deputy General Manager, Traffic Control Signal Technology (Beijing) Co,Ltd Wen Yilong is responsible for the technical innovation of the company's signalling system, leading the research and development of key products such as Perception-Based Train Autonomous Control System (PB-TACS) and Virtual Coupling.

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	Serena Shi General Manager, International Department I, Traffic Control Technology Co., Ltd Shi Zhu is a senior engineer and the general manager of the International Department I at Traffic Control Technology Co., Ltd. ("TCT"). She holds a Master's degree in Traffic Information Engineering and Control and has been with TCT since 2012, where she has assumed multiple roles. She is currently responsible for leading the company's international business across Hong Kong, Singapore, Australia, and other regions.
	Nie Lei Marketing Manager, International Department I, Traffic Control Technology Co., Ltd Nie Lei is a intermediate economist and holds a MBA degree of University of International Business and Economics ("UIBE"). From 2016 to 2024, he assumed the role of business manager in China Railway Construction Corporation International Limited ("CRCCI") and worked in Latin America for over 5 years, in charging of company business in Peru and Costa Rica. In 2025 he joined TCT and currently holds the position of marketing manager of Singapore.
	Zhang Wei (Tianjin GM) General Manager, Traffic Control Technology Equipment Zhang Wei, General Manager of Traffic Control Technology Equipment. Born in 1978, he holds a master's degree from Beijing Jiaotong University. He has served as General Manager of Traffic Control Technology Equipment since 2023. Prior to 2023, he held positions as Director of the Audit Department, IT Department and Safety & Quality Department.

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	Zhao Jianqiao (CdM Person In Charge) General Manager, Customer Success Center TRAFFIC CONTROL DEVOPS Upon graduation in 2012, Zhao Jianqiao has worked consecutively as a Project Technical Manager for key urban rail transit projects, including Beijing Metro Line 7, Hohhot Metro Line 1, Ningbo Metro Line 4 and Suzhou Metro Line 6. He led technical deployment and guaranteed the successful commissioning and official operation of more than 10 metro lines nationwide. He has taken part in the compilation of national, local and association standards, with 1 national standard, 1 local standard and 2 association standards officially released under his participation. Furthermore, he holds 4 authorized patents in relevant professional fields.
	Fan Yin (TCT VP, Join for Dinner) Vice President of Traffic Control Technology Co., Ltd Mr. Fan Ying is a Chinese national without foreign residency rights, born in January 1987. He is a Senior Engineer with a Master's degree in Traffic Information Engineering and Control from Beijing Jiaotong University, and a Bachelor's degree in Automation from the same university. From February 2012 to December 2022, he held various positions at Traffic Control Technology Co., Ltd., including Software Engineer at the R&D Center, Software Manager at the R&D Center, Product Manager for Heavy-Haul Train Control at the R&D Center, Assistant to the General Manager of the R&D Center, Deputy General Manager of the R&D Center, BU General Manager of the R&D Center, Assistant to the Company's General Manager, and Vice President of the company.

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5.2 CRRC Academy

5.2.1 Background



CRRC Industrial Research Institute Co., Ltd. (hereinafter referred to as the **CRRC Research Institute**) is a directly affiliated institution of CRRC. It serves as an extension, supporting, and service-oriented arm of CRRC's technological innovation management, as well as a public resource service platform for CRRC's technological innovation. The institute has been certified as both a "National High-Tech Enterprise" and a "Zhongguancun High-Tech Enterprise." Established in April 2013, it was registered in Fengtai District, Beijing, in August 2014 to begin substantive operations, and was renamed CRRC Industrial Research Institute Co., Ltd. in June 2015. In April 2020, it established a branch institute, "CRRC Industrial Research Institute (Qingdao) Co., Ltd.," within the National High-Speed Train Innovation Center in Qingdao.

CRRC Research Institute strives to secure CRRC's position as an industry leader. Driven by strategy and market demands, utilizing cross-border integration as a vehicle, and adopting a "**support + service**" model, the institute is dedicated to serving CRRC and its various subsidiaries. Its primary responsibilities include:

- Conducting research and consulting on macroeconomics, policies and regulations, market environments, development strategies, industrial growth, and management innovation for CRRC, as well as providing information intelligence services.
- Undertaking research on CRRC's technological development strategies and planning, and organizing the R&D of forward-looking, foundational, common, and core industry technologies, alongside the transfer and commercialization of technological achievements.
- Planning and building CRRC's product technology platforms.
- Building product R&D capabilities for CRRC's strategic products under the philosophy of "*exploring one generation ahead, pre-researching one generation ahead.*"
- Leading CRRC's technological R&D resources, and centrally managing tech-innovation organizations such as national-level R&D institutions (outside the Innovation Center), overseas R&D centers, branch research institutes, and various joint innovation bodies, while assuming investment responsibilities for CRRC's public technological innovation resources like the Innovation Center.
- Focusing on CRRC's technological development planning, technological strategies, and techno-economic research to become the core entity for technological information services, ultimately building a **CRRC Think Tank** to

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provide strong support for CRRC to achieve its "Dual Building and One Cultivation" strategic goals.

The CRRC Research Institute comprises 7 primary business and management service departments, which oversee 17 secondary offices. As of the end of December 2020, the institute had 92 regular employees, primarily consisting of four groups: management, economic research, technical research, and intelligence research personnel. Within the workforce, Ph.D. holders account for 27%, Master's holders account for 65%, and senior professional title holders account for 48% (including 12% with professor-level senior titles). Furthermore, 30% of the staff consist of top-tier talent, including recipients of the "Ten Thousand Talents Program," the State Council Special Allowance, the Mao Yisheng Science and Technology Award, the Zhan Tianyou Railway Science and Technology Award, as well as CRRC Chief Experts, CRRC Senior Experts, and CRRC International Talents.

5.2.2 Key Representatives

CRRC Academy	
	Xian Xiaoyu 咸晓雨 Deputy Director, Head of Department of Technology Research 技术研究部副部长 •负责技术研究部全面工作 Leads the Department of Technology Research • 统筹人工智能、算法及工业智能化 Oversees AI, algorithm innovation, and industrial intelligence programs • 推进高质量数据集及产业应用建设 Directs high-quality dataset and industrial AI Initiatives

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	Liu Qi 刘琦 Assistant Director of Department of Technology Research 技术研究部 部长助理 • 协助开展技术研究部管理工作 Assists in the management of the Department of Technology Research • 负责部门运营协调与资源统筹 Coordinates departmental operations and resources • 推动重点项目实施与跨部门协同 Supports key projects and cross-functional collaboration
	Liu Zhenbo 刘振波 Assistant Director of Department of Technology Research 技术研究部 部长助理 • 协助开展技术研究部管理工作 Assists in the management of the Department of Technology Research • 负责工业软件室、智能制造室建设与管理 Leads the Industrial Software Division and Intelligent Manufacturing Division • 负责技术部整体对外协同及数字化应用建设 Coordinates external collaboration and digital initiatives

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	Fan Yuming 樊玉明 Head of Intelligent Products Division, Department of Technology Research 技术研究部 智能产品室主任 - 负责智能产品室全面管理工作 Leads the Intelligent Products Division - 统筹人工智能产品研发与技术创新 Oversees AI product development and technology innovation - 推进高质量数据集、AI 平台及智能应用建设 Directs high-quality dataset development, AI platforms, and intelligent applications
	Chen Jiancheng 陈建成 Data Team Lead, Department of Technology Research 技术研究部 数据组负责人 - 负责高质量数据集相关项目组织与实施 Leads the implementation of high-quality dataset projects - 协调数据标准研究与建设工作 Coordinates data standards research and development activities - 推进数据资源建设与应用落地 Supports data resource development and application deployment

6 Talking Points

6.1. Opening Remarks with TCT

[Intros]

一龙, Serena, 谢谢大家抽出时间和我们交流, 我们终于见面了。我是万方, 是新加坡陆交局负责数字化转型的处长。这次来我们也一同带了两位数据方面的经理, 易蓁和杰聪, 和一起跟大家探讨数字化转型当中的挑战和数据的治理和应用。

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[Purpose]

此次行程我们最主要是想验收一下人型机器人的项目，了解一下交控在状态监测和人工智能领域的发展，并且看看有什么可以继续合作的地方。希望以后也可以多交流，多合作。交控有什么问题也可以提出来我们一起探讨。

6.2 Key Questions with TCT

<机器人>

轨道交通的运维工作往往涉及复杂环境和高安全要求，机器人技术的引入具有很大潜力。交控目前在哪些场景中部署了机器人，例如轨道巡检、设备维护、站台服务等？机器人与人工作业之间如何协同配合？在实际落地过程中遇到了哪些技术或管理层面的挑战？

<信号系统的状态监测>

信号系统是轨道交通安全运营的核心。交控信号状态监测系统的整体架构和核心功能是什么？系统如何应对网络安全威胁？是否有在不同城市或不同制式线路上的经验可以分享？

<AI 人工智能>

目前在哪些业务场景中应用了人工智能，例如故障预测、运行优化、客流分析等？AI 模型的训练数据来源是什么，数据质量如何保障？从研发到实际部署，AI 应用的落地周期大概是多久，主要瓶颈在哪里？

<Framework Agreement 长期合作>

陆交局的状态监测 传感器等数据, 会到一个平台上面. 邀请交控参与长期合作 - 比如模型的设计, 算法的开发. 欢迎积极交换意见, 提出问题.

6.3. Opening Remarks with CRRC Academy

[Intros]

咸部长，谢谢大家抽出时间和我们交流，我们终于见面了。我是万方，是新加坡陆交局负责数字化转型的处长。这次来我们也一同带了两位数据方面的经理，易蓁和杰聪，和一起跟大家探讨数字化转型当中的挑战和数据的治理和应用。

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[Purpose]

此次行程我们最主要是想了解一下研究院对数字化发展的研究，特别是高质量数据集的发展和在 AI 当中的应用，并且看看有什么可以合作和借鉴的地方。我们特别对于数据治理和人工智能这方面感兴趣。希望以后也可以多交流，多合作。研究院有什么问题也可以提出来我们一起探讨。

6.4 Key Questions with CRRC Academy

<陆交局 Introduction>

陆交局是新加坡负责规划、运营和监管陆路交通基础设施与服务的政府机构。近年来，陆交局积极推进数字化转型，通过数据和人工智能提升交通规划、运营效率及乘客出行体验。

在这一过程中，陆交局越来越重视建立**扎实的数据基础**，确保所收集和使用的**数据准确、治理规范、符合实际需求**，尤其是在探索人工智能应用的背景下。这也是我们此次希望向研究院学习**高质量数据集建设经验**的主要原因。我们也希望深入了解研究院的**数据治理框架**，以及这些经验是否可以借鉴应用于交通监管领域。

<数据治理与质量评估>

研究院如何定义“高质量”数据？主要从哪些维度评估数据质量，例如完整性、准确性、时效性、一致性等？如何在不同团队或数据来源之间统一执行数据质量标准？

<标准规范>

研究院是否建立了内部的数据集建设标准？这些标准是否与国家或国际框架接轨？对于来源不同、格式各异、质量参差不齐的数据，研究院是如何处理的？

<人工智能应用>

研究院基于这些高质量数据集开发了哪些人工智能模型或应用？从原始数据到可用于人工智能的数据集，主要面临哪些挑战？如何验证一个数据集是否真正适合用于人工智能训练或推理？

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<平台能力>

支撑数据集建设工作的数据平台基础设施是怎样的？如何管理数据血缘、版本控制和访问权限？

<合作机会>

陆交局的交通数据与研究院的科研能力是否存在互补空间？例如，在交通相关人工智能应用方面开展联合研究，或在数据质量评估方法论方面相互借鉴？

6.5 Other Detailed Questions

(A) Data Management & Cybersecurity

Data Management & Security

- 1) How is data managed across different transport systems within an integrated hub to ensure real-time decision-making and security?
- 2) What cybersecurity measures are necessary to protect critical infrastructure and prevent cyber threats in these large, interconnected hubs?

*Regulatory Compliance, Standards & Scalability **

- 3) What standards or regulations should be followed when designing and assessing high quality dataset?
- 4) Are there any local or regional regulations (like in Singapore) that should influence the dataset development or integration approach?
- 5) How do you ensure data completeness and traceability across the full data pipeline, from collection at source to consumption by AI models?
- 6) How do you manage data sharing and governance when working with external research partners or technology vendors?

(B) Technology Development & AI Applications

Research & Development Approach

- 7) What is your overall R&D philosophy — how do you decide which problems to prioritise, and how do you move from research to real-world deployment?
- 8) How do you validate and test new technologies before they are deployed in live transport operations? What does your testing and certification process look like?

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- 9) How do you keep pace with the rapid evolution of AI and machine learning technologies, and how do you decide which new capabilities to integrate into your products?

AI & Data Applications

- 10) In which operational scenarios have you deployed AI. For example, fault prediction, operations optimisation, or passenger flow analysis and what have been the most impactful outcomes?
- 11) What are the key challenges in moving from raw operational data to AI-ready datasets, and how do you address issues such as data noise, labelling, and class imbalance?
- 12) How do you measure the effectiveness of your AI deployments, what KPIs or evaluation frameworks do you use?
- 13) How do you handle model drift or degradation over time, and what processes are in place to retrain or update deployed models?

(C) System Integration & Deployment

Integration with Operator Systems

- 14) How do your solutions interface with existing IT and OT systems on the operator side, and what are the typical integration challenges you encounter?
- 15) How do you ensure interoperability across different rail systems, standards, and configurations when deploying your solutions in different cities or countries?
- 16) What does a typical deployment timeline look like, from initial engagement with an operator to full go-live, and where are the most common bottlenecks?

CdM & Signalling (TCT-specific)

- 17) What is the overall architecture and core functionality of the CdM system, and how does it compare to traditional signalling approaches in terms of reliability and flexibility?
- 18) How does the CdM system handle edge cases or failure scenarios, and what are the fallback mechanisms to ensure safety?

Robotics (TCT-specific)

- 19) In which maintenance or operational scenarios have robots been deployed, and how is the collaboration between robots and human workers managed on the ground?
- 20) What are the key technical challenges in deploying robots in live rail environments, such as dealing with dynamic conditions, safety certification, and integration with existing systems?

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(D) Collaboration & Knowledge Transfer

- 21) What models of collaboration do you typically engage in with transport operators. For example, joint pilots, co-development, or technology licensing?
- 22) How do you support knowledge transfer and capability building on the operator side when deploying new technologies?
- 23) Are there specific gaps or challenges in transport AI or data management that you are currently looking for partners to help address?
- 24) Given LTA's context as a transport regulator and operator in a dense urban environment, where do you see the most promising areas for collaboration?
- 25) How do you approach building long-term partnerships with overseas operators, and what has worked well in past international collaborations?